CHANDU PRASAD BHAIRAPU



AI Engineer

bhairapuchanduprasad@gmail.com (+49) 15560194877 Berlin • EU Blue Card linkedin github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building production-grade applied AI features, from agentic LLM workflows and document extraction to evaluation and human review, deployed on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in Python, full-stack development and machine learning.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning an applied AI system end-to-end in an early-stage startup: data pipelines ingesting unstructured regulatory and technical documents, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent LLM workflows with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel), with evaluations against an SAP-system-derived ground truth raising precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts review and approve knowledge-graph changes before they are applied (human review and approval).
- Designed and implemented a Model Context Protocol (MCP) integration exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, Vector Search, MCP, Postgres, MongoDB, Neo4j, FastAPI, React, Azure, Kubernetes, Docker**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to map their decision-making workflows and translated the requirements into an agentic AI application with LangGraph, deployed to AWS, that routes each request across vector database queries, general LLM inference and summarization.
- Developed a Python backend with FastAPI REST APIs and a React front-end, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK (infrastructure as code), following clean code practices; versioning in GitLab with CI/CD deployment.
- **Tech stack: Python, FastAPI, LangGraph, LangChain, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 | Leverkusen, Germany

- Automated LLM evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, LangChain, Hugging Face Transformers, PyTorch, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, CDK)**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Tender Document Extraction & Review | Python | Prefect | FastAPI | Postgres | React | Langfuse

- Prefect pipeline for scanned and digital construction tender PDFs: LLM and OCR extraction of line items, entity resolution across documents, a review UI linked to source pages, and evaluation with error analysis in Langfuse.

TECHNICAL SKILLSET

- **Applied AI & LLMs:** LLMs, agentic workflows (LangGraph, multi-agent), LangChain, LiteLLM, RAG, GraphRAG, vector DBs (Qdrant, Pinecone, Azure AI Search), prompt engineering, LLM evaluation (RAGAS), MCP, QLoRA, Claude Code
- **Full-stack & APIs:** Python, FastAPI, REST APIs, React (beginner), Postman, SQL, Postgres, MongoDB, Neo4j, PyTorch
- **Cloud & Operations:** AWS (CDK, Lambda, SQS, DynamoDB, EC2), Azure, Docker, Kubernetes, GitLab CI/CD, Agile
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India (Minor in CS & Engineering)

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles